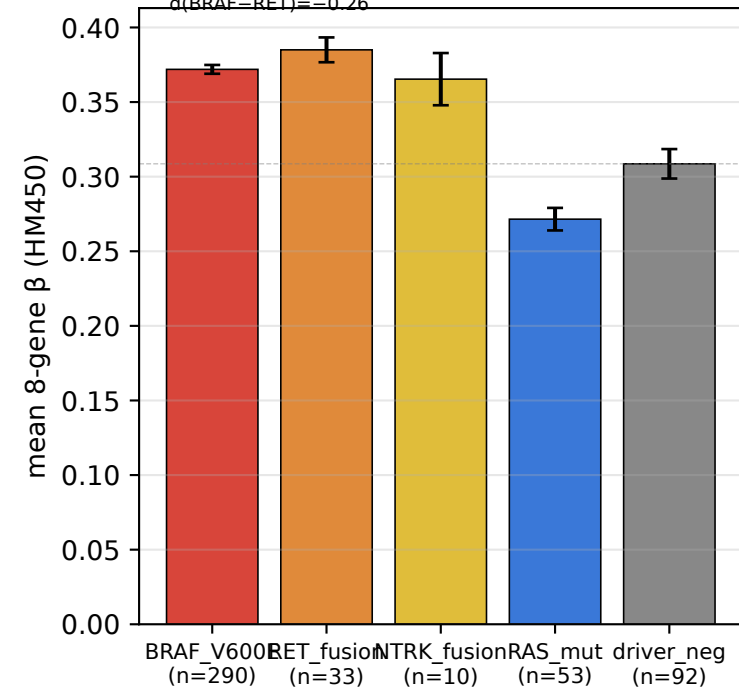
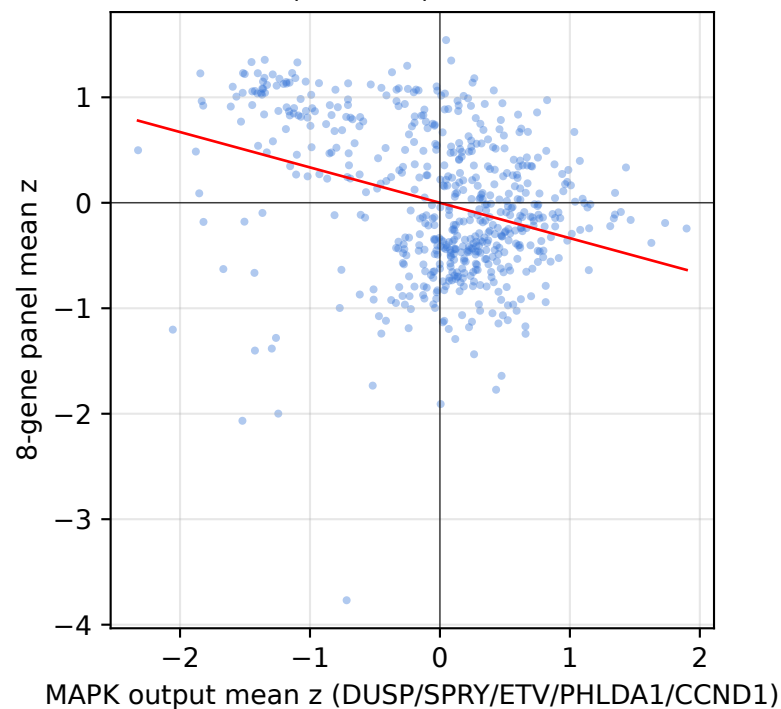


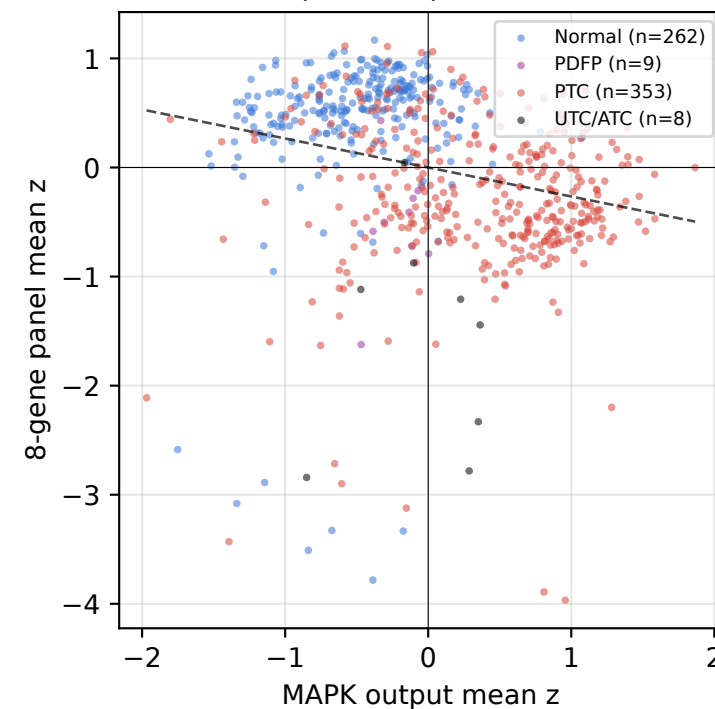
A. Driver-class × 8-gene methylation
MAPK-active (BRAF/RET/NTRK) hyper-methylate; RAS preserves
 $d(\text{BRAF}-\text{RET})=-0.26$



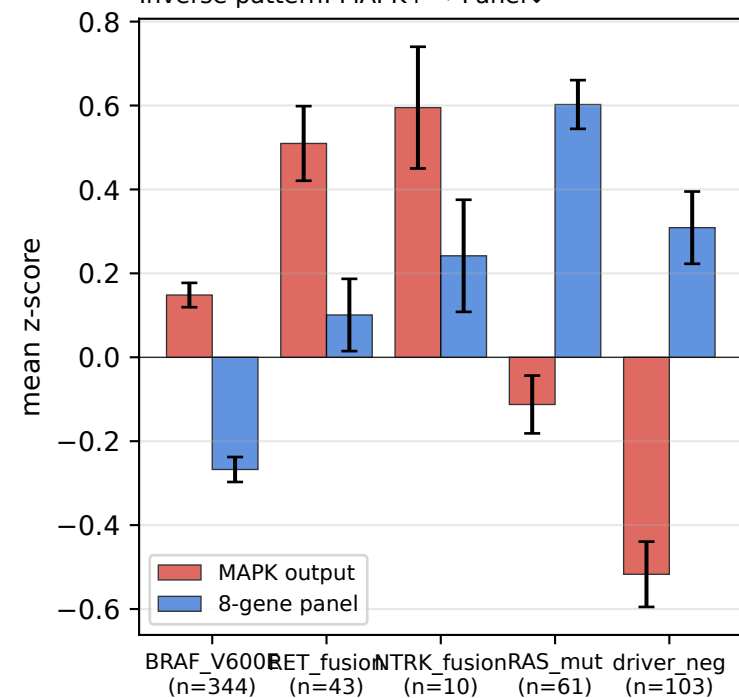
B. TCGA-THCA n=572
MAPK × Panel $\rho=-0.291$ $p=1.3e-12$



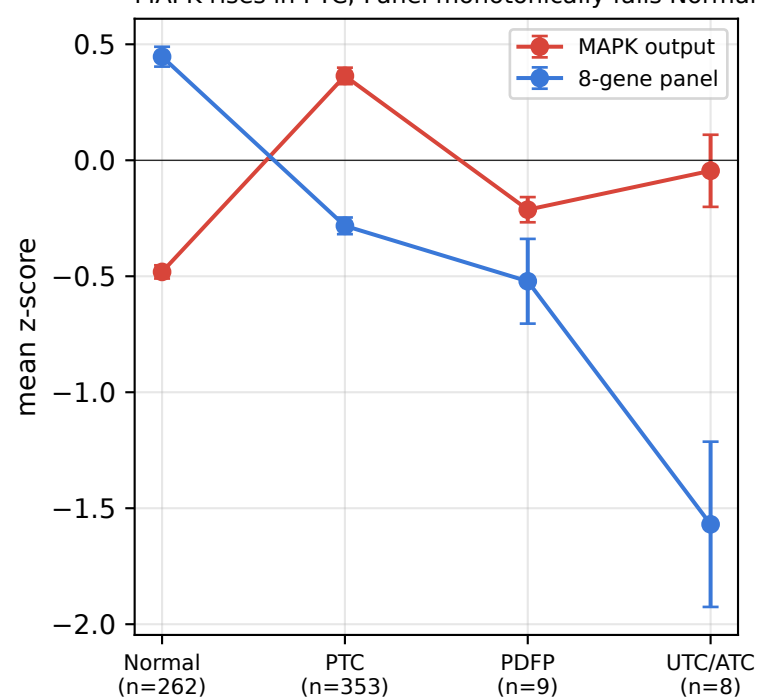
C. Lee GSE213647 n=632
MAPK × Panel $\rho=-0.395$ $p=4.8e-25$



D. Driver-class MAPK output vs 8-gene panel (TCGA)
Inverse pattern: MAPK $\uparrow$ → Panel $\downarrow$



E. Lee histology trajectory
MAPK rises in PTC; Panel monotonically falls Normal→ATC



F. MAPK → 8-gene silencing – mechanism

Driver class × HM450 methylation (TCGA n=478):
 BRAF V600E $\beta=0.37$, RET fusion $\beta=0.39$
 RAS mut $\beta=0.27$, driver-neg β baseline
 $d(\text{BRAF}-\text{RET})=-0.26$ (essentially equal)
 $d(\text{BRAF}-\text{RAS})=+1.97$, $d(\text{RET}-\text{RAS})=+2.17$
 → MAPK-output, NOT BRAF-specific

RNA cross-cohort MAPK × Panel:
 TCGA n=572: $\rho=-0.291$ $p=2.5e-12$
 Lee n=632: $\rho=-0.395$ $p=4.8e-25$
 → bi-cohort replication

Lee histology trajectory:
 Normal (n=262): MAPK=-0.48, Panel=+0.45
 PTC (n=353): MAPK=+0.36, Panel=-0.28
 ATC (n=8): Panel collapses to -1.57

Mechanism: MAPK pathway output (DUSP/SPRY/ETV transcriptional program) → DNA-methylation reprogramming at thyrocyte-lineage TF promoters → 8-gene silencing → DM1 phenotype.

Why RAS retains panel: RAS in thyroid is a weak MEK/ERK activator (vs strong PI3K) – known biology; explains FVPTC histology overlap.